RORY REARDON

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EDUCATION

California State Polytechnic University, Pomona Expected Graduation: December 2027
B.S. Computer Engineering | GPA: 3.15 | Dean's List, Spring 2026
Relevant Coursework: Intro to Microcontrollers, Digital Logic Design, Data Structures and Algorithms, Object-Oriented Programming, C for Engineers, Circuit Analysis I and II, Microelectronic Circuits
In Progress: Digital Design with Verilog, Operating Systems for Embedded Apps, Fundamentals of Cybersecurity, Signals and Systems

SKILLS

Languages: C, C++, Verilog, Python, C#, Java
Embedded & Hardware: STM32, PIC18F, FPGA, ESP32, SPI, I2C, UART, PWM, VGA, timing closure, PCB design
Tools: Vivado, Altium Designer, KiCad, MPLAB X, MATLAB, Blender, Git

ENGINEERING EXPERIENCE

Northrop Grumman Collaboration Project (NGCP) September 2026 – Present
Electrical Team, Unmanned Ground Vehicle | STM32, Altium
- Designing the fire suppression circuit for the vehicle's DAQ shield in Altium: a MOSFET solenoid driver with flyback diode, fuse, and safe-off pull-down, plus an arm interlock and servo and RC inputs
- Picked sensors with the payload team: tank pressure and INA226 valve current as requirements, water level and leak detection as optional, and cut flow, thermal, and distance sensing since firing is manual
- Studied the vehicle's inherited electronics (two STM32 Nucleo shields, backplane, power/BMS board, motor driver) from Altium schematics and documented how power and signals connect

PROJECTS

Blinkit: FPGA Digital Pet October 2026 – Present
Personal and class final project | Verilog, Nexys A7 (Artix-7), ESP32-S3
- Built a real-time 640x480 VGA face renderer in Verilog, driven by a 16-mood state machine with no soft processor
- Wrote SPI (ADXL362) and I2C (ADT7420) sensor drivers so shaking it or changing the temperature shifts its mood, plus a PWM sound engine for its voice
- Fixed a timing violation by pipelining the PWM sound engine, taking worst setup slack from -4.0 ns to +0.27 ns
- Went through 20 design revisions; now building a battery-powered ESP32-S3 keychain version with a round touchscreen and Bluetooth pairing

AI Cat Robot October 2026 – Present
Software Team, team of 8 | Jetson Orin NX
- Researching how robot pets like Sony aibo and Anki Cozmo choose behaviors to shape the cat's personality
- Assigned to train and run the cat's on-device AI model for voice interaction on a Jetson Orin NX

Autonomous Line-Following Ground Vehicle April 2026
Team of 4 | PIC18F46K22, Embedded C
- Wrote register-level C for 3-sensor IR line tracking with proportional steering and lost-line recovery
- Added an HC-SR04 ultrasonic sensor that stops the vehicle 5 inches from obstacles
- Fixed motor drift by tuning a separate PWM duty cycle for each motor through a TB6612FNG H-bridge
- Owned all mechanical design: modeled the chassis in Blender and 3D printed it in CR-PETG

WORK EXPERIENCE

In-N-Out Burger, Cerritos, CA | *Level 5 Associate* August 2024 – Present
- Keep orders accurate and moving during high-volume rushes while coordinating with a large team

LEADERSHIP & ACTIVITIES

Beta Tester, Independent App Projects | *QA and product feedback* September 2026 – Present
- Test new builds of two connected pre-release apps, a productivity app and a workout logger, reporting UI bugs like misaligned text and suggesting features like loading screens

Society of Hispanic Professional Engineers (SHPE) | *Member* 2022 – Present

Team Lead, C# Routing Project | *Team of 5* December 2025
- Split a greedy routing system into classes across the team, wrote the Temperature class, and led integration